Beyond Basic Strategy- Monte Carlo Simulation of Blackjack

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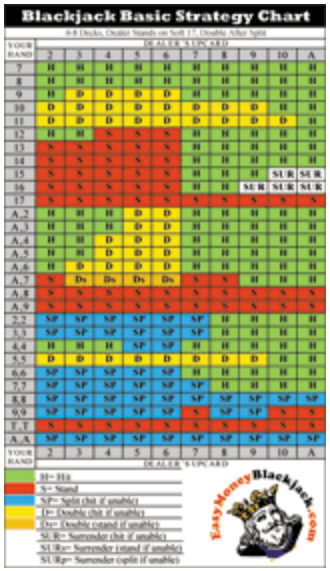
The Game of Blackjack

- Player and dealer draw cards to 21
- Hands are dependent
- Casino has ~0.5% edge

The Problem

- Do we want to only maximize expectation?
- Maybe we would rather maximize:
  - Number of hands played
  - Chance of doubling initial stake
- How do we do this?
  - Change playing strategy
  - Change bet sizing



Blackjack Basic Strategy Chart

4-8 Decks, Dealer Stands on Soft 17, Double After Split

YOUR HAND	DEALER SUPCARD 2	3	4	5	6	7	8	9	10	A
7	H	H	H	H	H	H	H	H	H	H
8	H	H	H	H	H	H	H	H	H	H
9	H	D	D	D	D	H	H	H	H	H
10	D	D	D	D	D	D	D	D	H	H
11	D	D	D	D	D	D	D	D	D	H
12	H	H	S	S	S	H	H	H	H	H
13	S	S	S	S	S	H	H	H	H	H
14	S	S	S	S	S	H	H	H	H	H
15	S	S	S	S	S	H	H	H	SUR	SUR
16	S	S	S	S	S	H	H	SUR	SUR	SUR
17	S	S	S	S	S	S	S	S	S	S
A,2	H	H	H	D	D	H	H	H	H	H
A,3	H	H	H	D	D	H	H	H	H	H
A,4	H	H	D	D	D	H	H	H	H	H
A,5	H	H	D	D	D	H	H	H	H	H
A,6	H	D	D	D	D	H	H	H	H	H
A,7	S	Ds	Ds	Ds	Ds	S	S	H	H	H
A,8	S	S	S	S	S	S	S	S	S	S
A,9	S	S	S	S	S	S	S	S	S	S
2,2	SP	SP	SP	SP	SP	SP	H	H	H	H
3,3	SP	SP	SP	SP	SP	SP	H	H	H	H
4,4	H	H	H	SP	SP	H	H	H	H	H
5,5	D	D	D	D	D	D	D	D	H	H
6,6	SP	SP	SP	SP	SP	H	H	H	H	H
7,7	SP	SP	SP	SP	SP	SP	H	H	H	H
8,8	SP	SP	SP	SP	SP	SP	SP	SP	SP	SP
9,9	SP	SP	SP	SP	SP	S	SP	SP	S	S
T,T	S	S	S	S	S	S	S	S	S	S
A,A	SP	SP	SP	SP	SP	SP	SP	SP	SP	SP
YOUR HAND	2	3	4	5	6	7	8	9	10	A

H= Hit
S= Stand
SP= Split (hit if unable)
D= Double (hit if unable)
Ds= Double (stand if unable)
SUR= Surrender (hit if unable)
SURs= Surrender (stand if unable)
SURp= Surrender (split if unable)

Methods

- Monte Carlo Simulation

Change Playing Strategy

Strategy	Buy-in	Bet Size	Goal Win Amount	Chance	Average Hands
Basic	$200	$10	$200	29%	371
Risk-Averse	$200	$10	$200	27%	381
No Extra Risk	$200	$10	$200	22%	366

Change Betting Strategy

Strategy	Buy-in	Average Bet Size	Goal Win Amount	Chance	Average Hands
After win ▲ After loss ▼	$300	$34	$300	42%	34
After win ▼ After loss ▲	$300	$40	$300	45%	41
Flat bet	$300	$35	$300	40%	132
After win ▲ After loss ▼ ▼	$300	$19	$300	38%	181
After win ▼ ▼ After loss ▲	$300	$22	$300	43%	175